

UNIVERSITY OF MINES AND TECHNOLOGY, TARKWA
FACULTY OF COMPUTING AND MATHEMATICAL
SCIENCES

DEPARTMENT OF MATHEMATICAL SCIENCES

A PROJECT REPORT ENTITLED

DEVELOP A MODEL TO PREDICT CUSTOMER CHURN
IN TELECOMMUNICATION SECTOR

BY

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SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENT FOR THE AWARD OF THE DEGREE
OF BACHELOR OF STATISTICAL DATA SCIENCE

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APRIL, 2026

DECLARATION

We declare that this project work is our own. It is being submitted for the degree of Bachelor of Statistical Data Science in the University of Mines and Technology (UMaT), Tarkwa. It has not been submitted for any degree or examination in any other university.

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DEDICATION

We dedicate this project to our families, lecturers, and friends for their support and encouragement throughout our academic journey.

ACKNOWLEDGMENT

First and foremost, we thank the Almighty God for His grace and protection throughout our studies.

We express our sincere gratitude to our supervisor, **Assoc. Prof. Peter Kwesi Nyarko**, for his guidance, encouragement, and support throughout this research.

We also appreciate our families and colleagues for their motivation and support. Finally, we must express our gratitude to most of our colleagues who helped us acquire more knowledge that improved the content of our project.

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Customer churn remains a critical challenge in the telecommunication sector, as losing customers directly affects revenue and increases acquisition costs. With the growth of large customer datasets, there is a need for effective analytical tools to predict churn behavior. Machine learning techniques have been widely adopted to improve prediction accuracy and support decision-making. However, selecting the most suitable model remains a challenge due to varying performance across algorithms. Therefore, this study focuses on developing and comparing predictive models to identify key factors influencing customer churn and support retention strategies.

1.2 Problem Statement

Customer churn continues to be a major challenge in the telecommunications sector because losing existing customers reduces revenue and increases the cost of acquiring new subscribers (Adebiyi, 2021). Despite the availability of large customer datasets, many telecommunication companies struggle to effectively utilize this data to predict and prevent churn accurately (Kumar and Ravi, 2021). Studies have shown that different machine learning algorithms produce varying levels of predictive accuracy, making it difficult to determine the most suitable model for churn prediction (Hossain, 2024).

Furthermore, customer behavior is often complex and non-linear, which reduces the effectiveness of single or traditional predictive models (Liu, 2022). Recent research emphasizes the importance of comparative analysis to identify the most accurate and reliable machine learning technique for predicting telecom customer churn (Sharma, 2023). In addition, there is limited empirical research focusing on churn prediction using machine learning within the telecommunications context of developing markets, highlighting a significant research gap (Mensah, 2025). Therefore, there is a need to develop and compare machine learning models capable of accurately predicting customer churn and identifying the key factors influencing customer switching behavior (Ahmed, 2023).

1.3 Objectives

The main objective of this study is to develop a predictive model for customer churn in Ghana's telecommunication sector using machine learning techniques.

Specific objectives include:

- i. Determine customer behavioral and service usage factors influencing churn.
- ii. Develop churn prediction models using Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine.
- iii. Evaluate the performance of the developed models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- iv. Provide insights that support data-driven customer retention strategies in the telecommunications sector.

1.4 Methods Used

This study would employ supervised machine learning techniques to predict customer churn. The selected models include:

- i. Logistic Regression
- ii. Decision Tree
- iii. Random Forest
- iv. Support Vector Machine

The models will be trained and evaluated using performance metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

1.5 Facilities and Resources Used

The following tools and resources were used for this project work:

- i. University internet facilities
- ii. University library resources

iii. Personal computer

iv. Python programming environment (Scikit-learn, Pandas, NumPy, Matplotlib)

1.6 Project Organization

This project is organized into five chapters. Chapter 1 is the introductory chapter which consist of the statement of the problem, project objectives, methods used and facilities needed for the project. Chapter 2 comprises of review of relevant literature in the study area. Chapter 3 gives a thorough discussion of the related concepts and methods needed to achieve the objectives. Chapter 4 includes model formulation, its analysis, simulation and results and Chapter 5 gives the conclusion and recommendation drawn from the project work and project organization.

CHAPTER 2

LITERATURE REVIEW

2.1 Conceptual Overview of Customer Churn

Customer churn has become a central research topic in telecommunications analytics due to its direct impact on revenue stability and long-term profitability (Saleh, 2023). Modern telecommunication markets are characterized by high competition and reduced switching barriers, increasing the importance of predictive churn management systems (Amin *et al.*, 2021). Recent literature emphasizes that predictive modelling frameworks are more effective than descriptive approaches in supporting proactive retention strategies (Liu, 2024).

Furthermore, the integration of explainable artificial intelligence techniques has strengthened churn analytics by improving model transparency and managerial trust (Liu, 2024). Contemporary research therefore views churn prediction not merely as a classification problem but as a decision-support system integrated into customer relationship management processes (Olaya, 2020).

2.2 Customer Churn

Customer churn refers to the termination of a customer’s service subscription within a defined period (Amin, 2021). In the telecommunications industry, churn prediction enables firms to identify subscribers with a high probability of defection before actual migration occurs (Saleh and Saha, 2023).

Recent empirical studies demonstrate that churn prediction systems significantly reduce revenue loss when integrated with targeted intervention mechanisms (Olaya, 2020). Machine learning-driven churn frameworks incorporate behavioural, transactional, and service-related variables to estimate churn probabilities with improved precision (Liu, 2024).

Additionally, telecommunication churn datasets are typically large-scale and multidimensional, requiring robust analytical models capable of handling complex nonlinear interactions (Amin, 2021).

2.3 Factors Influencing Telecommunication Customer Churn

Recent predictive modelling studies identify churn as a multidimensional outcome influenced by behavioural usage patterns, service experience, billing attributes, and customer engagement metrics (Saleh and Saha, 2023). Feature importance analyses using ensemble models confirm that multiple interrelated variables jointly influence churn probability (Liu, 2024).

2.3.1 Pricing and Tariffs

Pricing variables such as monthly charges, contract type, and payment method are consistently ranked among the strongest churn predictors in machine learning models (Amin, 2021). Empirical evidence shows that higher service costs relative to perceived value increase churn likelihood (Saleh and Saha, 2023). Furthermore, data-driven churn frameworks reveal that abrupt changes in billing amounts often precede subscriber defection (Liu, 2024).

2.3.2 Network Quality and Service Reliability

Network performance indicators, including service interruptions and usage decline, significantly influence churn probability (Sela and Saha, 2023). Studies applying ensemble classifiers show that subscribers experiencing inconsistent service quality exhibit higher predicted churn risk (Amin, 2021). Explainable AI analyses further demonstrate that reduced data consumption patterns frequently act as early churn signals (Liu, 2024).

2.3.3 Customer Service and Satisfaction

Customer interaction history, complaint frequency, and response time are important behavioral indicators of churn (Liu *et al.*, 2024). Predictive models incorporating service complaint variables achieve improved recall for minority churn classes (Amin *et al.*, 2021). Research also indicates that customer dissatisfaction, when unaddressed, significantly increases defection probability (Saleh and Saha, 2023).

2.3.4 Promotions and Brand Image

Targeted promotional strategies have been shown to reduce churn when applied to high-risk but persuadable customers (Olaya, 2020). Uplift modeling techniques demonstrate that generic retention offers are less effective than data-driven

personalized interventions (Olaya, 2020). Moreover, churn probability decreases when customers exhibit strong engagement responses to loyalty programs (Liu, 2024).

2.3.5 Billing Transparency and Accuracy

Billing consistency indicators improve predictive performance in churn detection systems (Amin, 2021). Research findings show that customers with irregular payment history or billing disputes display higher churn probabilities (Saleh and Saha, 2023). Feature attribution techniques confirm that billing-related variables contribute significantly to churn risk estimation (Liu, 2024).

2.3.6 Customer Engagement and Loyalty Programs

Customer engagement intensity, measured through usage frequency and service subscriptions, plays a significant role in churn modeling (Liu, 2024). Low engagement scores are strongly associated with increased churn probability in ensemble learning models (Saleh and Saha, 2023). Retention-focused studies emphasize that proactive engagement reduces churn more effectively than reactive strategies (Olaya, 2020).

2.3.7 Switching Incentives

Competitive telecom environments increase churn volatility, particularly when switching costs are low (Saleh and Saha, 2023). Empirical modeling studies show that churn probability rises in markets with aggressive promotional competition (Amin, 2021). Recent literature therefore highlights the importance of predictive monitoring in dynamic market conditions (Cerqueira, 2020).

2.4 Methodological Challenges in Churn Analytics

One of the most significant challenges in churn prediction is class imbalance, where churners represent a minority proportion of the dataset (Amin, 2021). Resampling techniques such as Synthetic Minority Over-Sampling Technique (SMOTE) and cost-sensitive learning significantly improve minority-class detection (Amin, 2021).

Another challenge is concept drift, which occurs when customer behavior patterns change over time (Cerqueira, 2020). Continuous retraining and adaptive learning frameworks are recommended to address this issue (Cerqueira, 2020). Temporal validation is also critical in telecommunication churn modeling, as random data splitting may inflate predictive performance (Liu, 2024). Chronological validation

strategies provide more realistic deployment evaluation (Liu, 2024).

Finally, accuracy alone is insufficient in imbalanced classification problems. Precision, recall, F1-score, and AUC-ROC are recommended for robust performance assessment (Saleh and Saha, 2023).

2.5 Statistical and Machine Learning Models for Churn Prediction

Logistic Regression remains widely applied due to interpretability, though its linear structure limits its ability to capture nonlinear interactions (Amin, 2021). Decision Trees provide rule-based classification but are prone to overfitting when used independently (Saleh and Saha, 2023). Random Forest improves predictive stability by aggregating multiple decision trees, thereby enhancing generalization performance (Saleh and Saha, 2023). Support Vector Machines demonstrate competitive binary classification performance in high-dimensional churn datasets (Amin, 2021).

Recent studies indicate that ensemble boosting algorithms outperform traditional models across recall and AUC metrics (Saleh and Saha, 2023). Explainable AI techniques such as Shapley Additive Explanations (SHAP) enhance interpretability by quantifying feature contributions to churn predictions (Liu, 2024).

2.6 Review of Recent Churn Prediction Studies

Recent studies have explored various approaches to improving customer churn prediction in the telecommunications sector. Amin (2021) demonstrated that integrating data certainty modeling with resampling techniques significantly enhances churn detection, particularly in imbalanced datasets. Saleh and Saha (2023) conducted a comparative analysis of ensemble machine learning algorithms and found that models such as Random Forest and boosting techniques consistently outperform traditional statistical methods in predictive accuracy. Olaya (2020) introduced uplift modeling approaches, which effectively distinguish between persuadable and non-persuadable customers, thereby improving the efficiency of retention strategies. Liu (2024) applied explainable artificial intelligence techniques to churn prediction and highlighted that improved model interpretability strengthens managerial decision-making. Furthermore, Cerqueira (2020) emphasized the importance of adaptive modeling frameworks in addressing concept drift and evolving data patterns in predictive systems.

2.7 Related Works

Several recent studies between 2020 and 2025 have examined telecommunication customer churn using advanced machine learning and analytical frameworks. Amin (2021) developed a churn prediction model using data certainty-based classification techniques combined with resampling methods to address class imbalance. Their study demonstrated that applying SMOTE significantly improved minority-class detection and enhanced F1-score performance. The strength of this study lies in its methodological rigor in handling imbalanced telecommunication datasets. However, the study primarily focused on predictive accuracy and did not integrate managerial decision frameworks for targeted retention (Amin, 2021). Saleh and Saha (2023) conducted a comparative analysis of ensemble machine learning models, including Random Forest, Gradient Boosting, and Support Vector Machines. Their findings revealed that ensemble-based classifiers consistently outperformed traditional statistical models in terms of precision, recall, and AUC-ROC metrics. A key contribution of their work is the comprehensive benchmarking of algorithms under identical validation settings. Nevertheless, the study placed limited emphasis on explainability and business interpretability of model outputs (Saleh and Saha, 2023). Liu (2024) proposed an explainable machine learning framework for telecom churn prediction using SHAP-based feature attribution methods. Their results showed that integrating explainable AI techniques significantly enhances model transparency and managerial trust. The study’s major strength is its focus on interpretability alongside predictive performance.

However, the research did not incorporate uplift modeling or retention optimization strategies (Liu, 2024). Olaya *et al.*(2020) introduced uplift modeling techniques for churn prevention, distinguishing between persuadable and non-persuadable customers. Their findings demonstrated that uplift-based targeting improves retention campaign profitability compared to traditional risk-based ranking models. The strength of this work lies in its integration of predictive analytics with actionable business decision-making. However, the study did not extensively evaluate model robustness under class imbalance conditions (Olaya, 2020). Cerqueira *et al.*(2020) examined concept drift adaptation in predictive systems and highlighted the importance of continuous model retraining in dynamic environments. Their findings emphasized that failure to account for evolving data distributions leads to deterioration in predictive performance over time. While their contribution is highly relevant for operational deployment, the study focused broadly on time-series forecasting rather than telecommunication specific churn datasets (Cerqueira, 2020).

Collectively, these studies demonstrate that recent churn research emphasizes ensemble

learning, imbalance correction, interpretability, and adaptive modeling. However, gaps remain in integrating predictive performance, explainability, uplift targeting, and contextual telecom deployment within a unified modeling framework (Amin *et al.*, 2021; Liu *et al.*, 2024; Saleh and Saha, 2023).

2.8 Research Gaps

Despite significant advancements in telecommunication churn prediction between 2020 and 2025, several important research gaps remain evident in the literature.

Firstly, although recent studies demonstrate high predictive accuracy using ensemble and boosting algorithms, many focus primarily on statistical performance metrics without adequately integrating predictive results into actionable retention strategies (Saleh and Saha, 2023). While models may achieve strong AUC and F1-scores, limited attention is given to how predictions translate into operational customer retention frameworks (Liu, 2024). This disconnect reduces the managerial applicability of many churn prediction systems.

Secondly, although explainable artificial intelligence techniques such as SHAP have improved interpretability, few studies combine explainability with intervention-based modeling approaches such as uplift modeling (Liu and Olaya, 2020). Most predictive systems rank customers based on churn probability without distinguishing between persuadable customers and those unlikely to respond to retention efforts (Olaya, 2020). This gap limits cost-effective targeting strategies in competitive telecommunication environments.

Thirdly, class imbalance remains a widely acknowledged challenge, yet not all predictive studies implement robust imbalance correction techniques (Amin, 2021). Some research evaluates models using accuracy-based metrics that may overestimate performance in skewed datasets (Saleh and Saha, 2023). There is therefore a need for consistent adoption of resampling techniques and imbalance-sensitive evaluation metrics in churn modeling frameworks.

Fourthly, the issue of concept drift and evolving customer behavior remains underexplored in many telecom churn applications (Cerqueira, 2020). While theoretical discussions emphasize adaptive retraining systems, practical implementation of continuous monitoring pipelines in telecommunication churn environments remains limited (Cerqueira, 2020). Failure to address behavioral shifts over time may reduce long-term model reliability.

Again, many recent churn studies rely on publicly available benchmark datasets, which may not fully capture the behavioral and competitive dynamics present in developing telecommunication markets (Amin, 2021). Consequently, contextualized modeling approaches tailored to specific telecommunication environments remain limited (Saleh and Saha, 2023). This gap highlights the need for comparative modeling studies that reflect real-world competitive conditions.

Finally, although ensemble methods dominate predictive performance rankings, comparative studies evaluating interpretability, computational efficiency, and deployment feasibility alongside accuracy remain insufficient (Liu, 2024). Practical telecom systems require a balance between predictive strength, transparency, and operational scalability.

Addressing these gaps provides justification for the present study, which seeks to develop and comparatively evaluate predictive models while emphasizing interpretability, imbalance handling, and deployment-oriented evaluation within a competitive telecommunication context.

CHAPTER 3

METHODS USED

3.1 Research Design and Data Source Description

This study adopts a quantitative predictive modeling research design. The objective is to develop and comparatively evaluate machine learning models for predicting customer churn in the telecommunications sector.

Due to privacy and confidentiality constraints associated with real telecommunication subscriber data, the study utilizes a structured telecommunication churn dataset inspired by industry-standard datasets such as the IBM Telecommunication Customer Churn dataset. The dataset is curated to reflect telecommunication service attributes relevant to competitive market conditions.

The dataset contains customer-level observations, where each row represents an individual subscriber and each column represents a predictor variable describing demographic characteristics, service usage behavior, billing patterns, and engagement indicators. The target variable is a binary classification outcome:

- i. Churn = 1 (Customer leaves service)
- ii. Churn = 0 (Customer stays)

The probability of churn is expressed as Equation (3.1)

$$P(Y) = 1 \mid X \tag{3.1}$$

where:

Y = customer churn status and X = vector of explanatory variables

3.2 Data Collection and Variables

The data used in this study consists of structured telecommunication customer records containing both numerical and categorical variables. These variables were carefully selected based on their relevance to customer churn behavior. The dataset captures customer demographics, account characteristics, service usage patterns, billing information, and the target churn outcome.

3.2.1 Demographic Variables

Demographic variables refer to the personal and household characteristics of customers that may influence their likelihood of churning. Although demographic features alone may not directly cause churn, they provide important contextual information that can improve model performance when combined with behavioral and service-related variables.

The demographic variables considered in this study include:

- i. Gender: Indicates whether the customer is male or female. This variable helps assess whether churn behavior differs across gender groups.
- ii. Senior Citizen Status: A binary indicator showing whether a customer is a senior citizen or not. Senior citizens may have different service needs and sensitivity to pricing or service disruptions.
- iii. Dependents: Indicates whether the customer has dependents. Customers with dependents may exhibit higher service stability due to family or household commitments.

These variables are encoded into numerical format during preprocessing to ensure compatibility with machine learning algorithms. Their inclusion helps the model capture underlying demographic patterns that may indirectly influence churn behavior.

3.2.2 Account Information

Account information describes the contractual and subscription-related characteristics of customers. These variables are particularly important because they reflect the stability and duration of the customer-service provider relationship.

The variables include:

- i. Tenure: Represents the total duration (in months) that a customer has stayed with the telecom provider. Customers with longer tenure are generally less likely to churn due to established loyalty.
- ii. Contract Type: Refers to the type of subscription agreement, such as month-to-month, one-year, or two-year contracts. Short-term contracts often show higher churn rates due to flexibility in switching.

- iii. Payment Method: Indicates how the customer pays their bills (e.g., electronic check, bank transfer, or credit card). Certain payment methods may be associated with different churn risks due to convenience or financial behavior patterns.

These variables provide strong predictive power because they directly reflect customer commitment and contractual obligations with the service provider.

3.2.3 Service Usage Variables

Service usage variables capture the extent and nature of services utilized by customers. These variables are critical in churn prediction as they reflect customer engagement and satisfaction with the telecom services.

The variables include:

- i. Internet Service: Indicates whether the customer subscribes to internet services such as DSL or fiber optic.
- ii. Online Security: Shows whether the customer has activated security services, which may influence trust and perceived value.
- iii. Streaming Services: Represents usage of entertainment-related services such as streaming TV or movies.
- iv. Multiple Lines: Indicates whether a customer has more than one telephone line under the same account.

These variables help in understanding how actively customers engage with different service offerings. Low service utilization may signal disengagement, which is often an early indicator of potential churn.

3.2.4 Billing Variables

Billing variables represent the financial interaction between the customer and the service provider. These variables are highly influential in churn prediction because pricing dissatisfaction is one of the most common reasons for customer departure.

The billing variables include:

- i. Monthly Charges: Represents the amount billed to the customer each month for services rendered. High monthly charges relative to perceived service value may increase churn risk.
- ii. Total Charges: Represents the cumulative amount charged to the customer over the entire subscription period. This variable reflects long-term financial commitment.

3.2.5 Target Variable

- i. Churn (Yes/No)

These variables were selected based on empirical evidence from recent churn literature highlighting pricing, engagement, and usage behavior as dominant predictors.

3.3 Data Preprocessing and Feature Engineering

Data preprocessing and feature engineering are critical stages in the development of robust and reliable machine learning models. These processes ensure that the dataset is clean, consistent, and suitable for analysis, while also enhancing the predictive power of the models by transforming raw data into meaningful features. In this study, several preprocessing techniques were applied to prepare the dataset for modeling.

3.3.1 Data Cleaning

Data cleaning was performed to improve the quality and integrity of the dataset. This involved identifying and handling missing values using appropriate imputation techniques such as mean or mode substitution, depending on the variable type. Inconsistent data entries and incorrect data types were corrected to ensure uniformity across the dataset. Additionally, duplicate records were detected and removed to prevent bias and redundancy in the analysis.

3.3.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to gain insights into the structure and distribution of the data. Statistical summaries and visualization techniques such as histograms, box-plots, and bar charts were used to understand variable distributions, detect outliers, and identify potential relationships between variables. EDA also helped

in assessing class imbalance in the target variable (churn) and guided subsequent preprocessing decisions.

3.3.3 Feature Engineering

Feature engineering was carried out to improve model performance by creating new variables that better represent customer behavior. For instance, tenure was grouped into categories such as short-term, medium-term, and long-term customers to capture retention patterns. Additional features such as average monthly spending and total service subscriptions were derived to reflect customer engagement and usage intensity. These transformations enhance the model's ability to detect hidden patterns in the data.

3.3.4 Encoding and Scaling

Since machine learning algorithms require numerical input, categorical variables were transformed into numerical format using One-Hot Encoding. This method ensures that categorical data is represented without introducing ordinal relationships. Furthermore, numerical variables were standardized using scaling techniques to ensure that all features contribute equally to the model. Standardization was performed using the Z-score transformation to normalize the data distribution.

3.3.5 Handling Class Imbalance

Customer churn datasets are often characterized by class imbalance, where the number of non-churn customers significantly exceeds churn customers. To address this issue, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to generate synthetic examples of the minority class. This approach improves the model's ability to correctly identify churn cases and enhances overall predictive performance, particularly in recall and F1-score metrics.

3.4 Model Development and Selection

Model development and selection are essential stages in building an effective predictive system for customer churn. This phase involves selecting appropriate machine learning algorithms, training the models on prepared data, and optimizing their performance to achieve reliable predictions. In this study, multiple supervised learning algorithms were implemented and compared to determine the most suitable model for churn prediction.

3.4.1 Data Splitting

To ensure an unbiased evaluation of model performance, the dataset was divided into three subsets: training, validation, and testing sets. The training set (70%) was used to build and train the models, while the validation set (15%) was used to fine-tune model parameters and prevent overfitting. The test set (15%) was reserved for final performance evaluation to assess how well the models generalize to unseen data.

3.4.2 Model Training

The study employed four supervised machine learning algorithms: Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM). Logistic Regression was selected due to its simplicity and interpretability in binary classification problems. Decision Tree was used for its ability to model nonlinear relationships and provide rule-based decision structures. Random Forest, an ensemble learning method, was implemented to improve prediction accuracy and reduce overfitting by aggregating multiple decision trees. Support Vector Machine was included due to its effectiveness in handling high-dimensional data and complex classification boundaries. All models were implemented using Python libraries such as Scikit-learn, NumPy, and Pandas.

3.4.3 Hyperparameter Tuning

To optimize model performance, hyperparameter tuning was conducted using Grid Search Cross-Validation. This process involves systematically testing different combinations of parameter values to identify the optimal configuration for each model. For instance, Random Forest parameters such as the number of trees and maximum depth were tuned, while Decision Tree parameters such as minimum samples split were adjusted. For SVM, kernel types and regularization parameters were optimized.

3.5 Model Evaluation Metrics

Model evaluation metrics are used to assess the performance and effectiveness of the developed machine learning models. Since customer churn prediction is an imbalanced classification problem, relying on a single metric such as accuracy may be misleading. Therefore, multiple evaluation metrics are used to provide a comprehensive assessment of model performance. They are as follows:

- i. Accuracy: measures the proportion of correctly classified instances out of the total

observations. The accuracy evaluation metric is given as equation (3.2)

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3.2)$$

Where TP represent True Positive, TN represent True Negative, FP represent False Positive and FN represent False Negative.

ii. Precision: measures the proportion of correctly predicted churn cases out of all predicted churn cases. The precision evaluation metric is given as Equation (3.3)

$$Precision = \frac{TP}{TP + FP} \quad (3.3)$$

iii. Recall (Sensitivity): measures the proportion of actual churn cases correctly identified. The recall evaluation is given as Equation (3.4)

$$Recall = \frac{TP}{TP + FN} \quad (3.4)$$

iv. F1-Score: represents the harmonic mean of precision and recall. The F1-Score evaluation is given as Equation (3.5)

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (3.5)$$

v. Area Under the ROC Curve (AUC-ROC)

AUC evaluates the model's ability to distinguish between churners and non-churners across thresholds. Precision, Recall, F1-score, and AUC are prioritized over accuracy due to class imbalance. A comparative performance table is presented to determine the best-performing model. This is represented as Table 3.1:

Table 3.1: Performance Evaluation Metrics

Metric	Definition	Formula
Accuracy	Overall correctness of the model	$\frac{TP+TN}{TP+TN+FP+FN}$
Precision	Correctness of positive predictions	$\frac{TP}{TP+FP}$
Recall	Ability to detect actual churn cases	$\frac{TP}{TP+FN}$
F1-Score	Balance between precision and recall	$2 \times \frac{Precision \times Recall}{Precision+Recall}$
ROC-AUC	Ability to distinguish between classes	Area under ROC curve

3.6 Ethical Considerations

Ethical considerations are essential in ensuring responsible research practices. In this study, data privacy and confidentiality were strictly maintained, and no personally identifiable information was used. The dataset was anonymized and used solely for academic purposes.

Additionally, care was taken to ensure that the developed models do not introduce bias or discrimination. The results are intended to support decision-making and improve customer retention strategies, not to unfairly target individuals. The study adheres to principles of transparency, reproducibility, and accountability by clearly documenting all methodologies and processes used.

REFERENCES

- Adebisi, A.A., Adewumi, A.O. and Ayo, C.K. (2021), “Comparison of data mining techniques for customer churn prediction”, *Expert Systems with Applications*, Vol. 39, No. 3, pp. 250–259.
- Ahmed, M., Mahmood, A.N. and Islam, M.R. (2023), “A survey of machine learning techniques for customer churn prediction”, *International Journal of Data Science*, Vol. 8, No. 2, pp. 112–130.
- Amin, A., Anwar, S., Adnan, A., Nawaz, M. and Howard, N. (2021), “Customer churn prediction in the telecommunication sector using a rough set approach”, *Neurocomputing*, Vol. 237, pp. 242–254.
- Cerqueira, V., Torgo, L. and Mozetič, I. (2020), “Evaluating time series forecasting models under concept drift”, *Machine Learning*, Vol. 109, No. 7, pp. 1393–1419.
- Chandar, M., Laha, A. and Krishna, P. (2021), “Modeling customer churn in telecom industry using logistic regression”, *International Journal of Engineering Research*, Vol. 10, No. 4, pp. 200–208.
- Han, J., Kamber, M. and Pei, J. (2021), *Data Mining: Concepts and Techniques*, 4th ed., Morgan Kaufmann, Burlington.
- Hossain, M.S. (2024), “Machine learning approaches for telecom churn prediction: A comparative study”, *Journal of Big Data Analytics*, Vol. 11, No. 1, pp. 45–60.
- Huang, B., Kechadi, T. and Buckley, B. (2020), “Customer churn prediction in telecommunications”, *Expert Systems with Applications*, Vol. 39, No. 1, pp. 1414–1425.
- Idris, A., Khan, A. and Lee, Y.S. (2020), “Intelligent churn prediction in telecom”, *Applied Intelligence*, Vol. 39, No. 3, pp. 659–672.
- James, G., Witten, D., Hastie, T. and Tibshirani, R. (2021), *An Introduction to Statistical Learning*, 2nd ed., Springer, New York.

- Kumar, V. and Ravi, V. (2021), “Applications of data mining in financial domain”, *Knowledge-Based Systems*, Vol. 114, pp. 128–147.
- Li, H. (2022), “Customer churn prediction using deep learning techniques”, *IEEE Access*, Vol. 10, pp. 34567–34578.
- Liu, Y. (2024), “Explainable artificial intelligence for customer churn prediction”, *Expert Systems with Applications*, Vol. 210, p. 118435.
- Mensah, K. (2025), “Telecommunication customer churn prediction in developing economies”, *African Journal of Data Science*, Vol. 5, No. 1, pp. 22–38.
- Ngai, E.W.T., Xiu, L. and Chau, D.C.K. (2020), “Application of data mining techniques in CRM”, *Expert Systems with Applications*, Vol. 36, No. 2, pp. 2592–2602.
- Olaya, D., Vásquez, J. and Maldonado, S. (2020), “Uplift modeling for preventing customer churn”, *Expert Systems with Applications*, Vol. 160, p. 113659.
- Saleh, M. and Saha, S. (2023), “Ensemble machine learning models for customer churn prediction”, *Journal of Data Mining*, Vol. 37, No. 2, pp. 789–812.
- Sharma, A. (2023), “Comparative analysis of machine learning algorithms for churn prediction”, *International Journal of Computer Applications*, Vol. 185, No. 12, pp. 15–22.
- Verbeke, W., Dejaeger, K., Martens, D., Hur, J. and Baesens, B. (2020), “New insights into churn prediction”, *European Journal of Operational Research*, Vol. 218, No. 1, pp. 211–229.